

Digital library case study

Summary

A "Digital Library" company has tasked Zühlke with developing a cloud-based solution for their already existing on-premise 3-tier application. Current solution has a frontend application written in React and backend as microservices, written in Java. Microservices are decoupled and are communicating via message broker with each other. Data that's being used is stored in a local database installed on their servers.

We need to do migration of this application to the cloud with minimal costs and maintenance effort. And after migration we can also try to utilize as many cloud services/benefits as possible.

The system shall be implemented on AWS cloud platform.

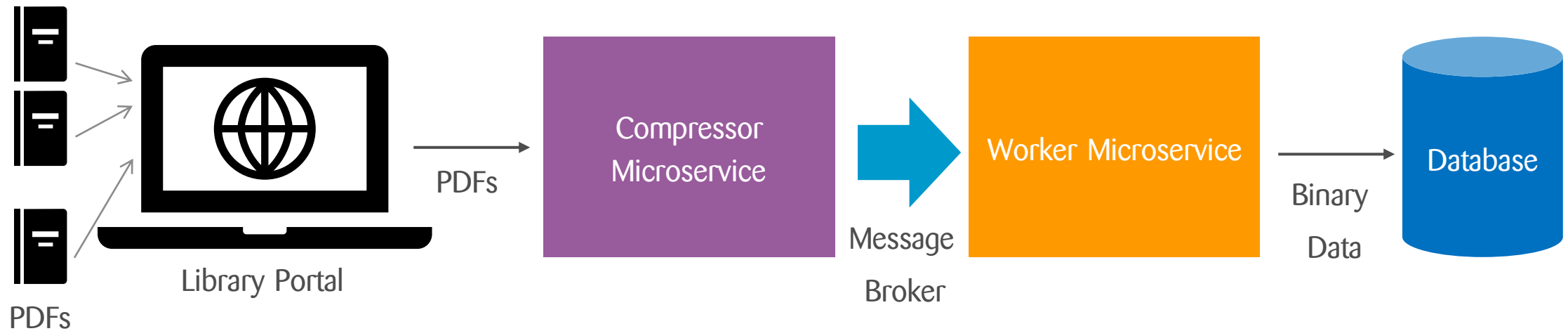
Apart from developing the solution, Zühlke is also contracted to support the customer in production with continuously evolving the solution, operations for the system components in the cloud and 3rd level support for the components in Zühlke's responsibility.

Some things to bear in mind

- Try and think about all the things you need to consider when designing a system. How are these aspects supported by your design?
- We aren't looking for any specific technology solution (Fly your Imagination). Work with what you are comfortable with.
- If you have any questions or need clarification on anything, please ask. If you need to use Google that's fine.
- This is a paper exercise, but we don't expect perfect diagrams and documentation. Talking us through your decision making is preferable to capturing everything perfectly in a diagram.
- **We will expect you to share your screen while you draw your flow of thoughts and feel free to choose the Drawing tool you wish to.** Some examples of drawing tools are draw.io or Paint.

Application flow

- The important component of the system is "Library Portal" which is an application that company employees use to upload new books as PDF files. The employee can go to portal web app, upload the book and submit it.
- After the submit, the data is sent to "Compressor" microservice that compresses the book and sends it to message broker.
- The "Worker" microservice takes the data from broker and saves it to the database as plain binary data.



The exercise needs

As an engineer with experience of developing and running live systems, you are included in the Zühlke product team from the very start to help them develop a DevOps mindset and, given this context, advise them on how to:

- Design a cloud-based architecture of web application covering the required components: What cloud services and patterns would you recommend ?
- Model a proper development, test and production environment for the whole solution
- Design a CI / CD architecture with tools of choice to promote application through environments

Please be prepared for a session of about 60 min (25 min presentation and 35 min discussion) in which we discuss your recommendations for each of the 3 questions. We would like to ask you to provide some **diagrams** providing an overview over the system architecture as well the infrastructure architecture. Try to consider some security and operational aspects.

You can assume that you have a technical audience (a typical Zühlke development team), so please be as concrete as it is possible for you. If you need anything or have questions/remarks, do not hesitate asking us.

Cloud strategy / non-functional requirements

- **Infrastructure as a code** is a de facto standard
- Systems should be **observable** at every handshake
- **Scalability** is part of the business model:
 - Solution should be cost effective
 - Infrastructure should scale up during high volumes, and scale down during low volumes and be performant while doing so
- Quality insurance is business critical